

Home Assignment. PySpark Medallion Pipeline. Tel Aviv Municipality Data

Context

You will build a small data platform flow using two municipal data sources, then produce a business output for 2023 compensation costs.

Data sources

1. Street closures history (CSV)

You will receive a CSV file that contains street closure events across many years.

Important: each row contains two street names, representing a closure segment between two streets.

https://storage.googleapis.com/test_onedatai/rechov_sagur.csv

2. Businesses table (API, JSON)

ArcGIS REST endpoint (query returns features with attributes and geometry).

API URL:

https://gisn.tel-aviv.gov.il/arcgis/rest/services/IView2/MapServer/925/query?where=1%3D1&outFields=*&f=json

Goal

Build a **Medallion Architecture pipeline** in PySpark/Python, then produce compensation analytics for 2023 based on street closures and business locations.

Cleaning and normalization

Create clean and consistent datasets from both sources.

- **Street closures data**
 - Parse closure start and end times to proper dates.
 - If end date is missing, use start date.
 - Ensure both streets in each row can be matched independently.
- **Business data**
 - Flatten API JSON into a table structure.
 - Convert area and dates to proper numeric and timestamp types.
 - Remove duplicate businesses using business id, keeping the latest record.

Business Rules for Compensation

1. A business is eligible for compensation on days when its street appears in closure data.
2. Compensation is calculated once per business per day.
3. If a street is marked as closed, the business receives at least 100 ILS per day foreach 1 "shetach".
(For example 25 shetach for 2 days will get $2 \times 25 \times 100 = 5,000$ ILS)
4. The Maximum receives for each business is **10,000 ILS per day**.
5. Each closure row contains two streets. Both must be considered affected.
6. Compensation is calculated only for the year 2023.

You will implement at least one compensation model, and you will also calculate "cost per area" and at least one additional metric.

Final Business Outputs

1. Annual compensation totals per business for 2023.
2. Annual compensation totals aggregated per street for 2023.

Deliverables

- Code - Provide a runnable PySpark pipeline implemented as a notebook or Python package.
- Data Outputs-Provide the resulting datasets generated by the pipeline, including cleaned data and final business outputs.

Bonus Tasks

1. Implement the solution using Databricks.
2. Add a scheduler to automate pipeline execution.
3. Build a dashboard to present key metrics and results.
4. Produce additional insights derived from the data analysis.